Cpts 317. 11/14.

Prob 1. (follow Nov. 2 lecture)

Prob 2. (follow Nov. 4 lecture).

Alg:

① $\delta(p, a, A) \ni (q, \Lambda)$

$\Rightarrow \quad [p, A, q] \longrightarrow a$

② $\delta(p, a, A) \ni (p_1, \gamma)$ with $\gamma \neq \Lambda$,

let $\gamma = B_1 \ldots B_m$

$\Rightarrow$

$[p, A, q] \longrightarrow a[p, B_1, p_2] \ldots [p_{m-1}, B_m, p_m]$

with $p_m = q$.

③ · Initial rule:

$$S \longrightarrow [q_0, Z_0, q_f].$$

Exaple: I have instruction

$$\delta(q_0, 0, Z_0) \ni (q_0, 0Z_0)$$

$$\Rightarrow [q_0, Z_0, *_1] \rightarrow 0[q_0, 0, *_2][*_2, Z_0, *_1]$$

where $*_1, *_2 \in Q$.

Ex: $\delta(q_0, *_1, *_2) = \{(q_0, *_1 *_2)\}$ with $*_1, *_2 \in \{0,1\}$.

replacement:

$\delta(q_0, a_1, a_2) \ni (q_0, a_1 a_2)$ with $a_1, a_2 \in \{0,1\}$

$\Rightarrow [q_0, a_2, *_1] \to a_1 [q_0, a_1, *_2] [*_2, a_2, *_1]$,

with $*_1, *_2 \in Q$,
$a_1, a_2 \in \{0,1\}$

$Q = \{q_0, q_{even}, q_{odd}, q_2\}$

Prob 3. If L_1, L_2 are reg, then

$$\mathcal{L}_3 = \{ x x^r : x \in L_1 , \ x^r \in L_2 \} \text{ is}$$

c.f.

Proof. ~~Ob:~~

$$\cancel{\mathcal{L}_3 = \{ x x^r : x \in \Sigma^* \} \cap L_1 \cdot L_2}$$

Define $\mathcal{L}_4 = \{ x \# x^r : x \in L_1 , \ x^r \in L_2 \}$

Then, $\mathcal{L}_4 = \{ x \# x^r : x \in \Sigma^* \} \cap L_1 \cdot \{\#\} \cdot L_2$.

Clealy, $\mathcal{L}_4$ is c.f. since it's intersection between a c.f. lang with a reg lang.

The result follows by replacing '#' with λ in a grammar generating the $\mathcal{L}_4$.

Prob.4. using P.L.

Turing machine.

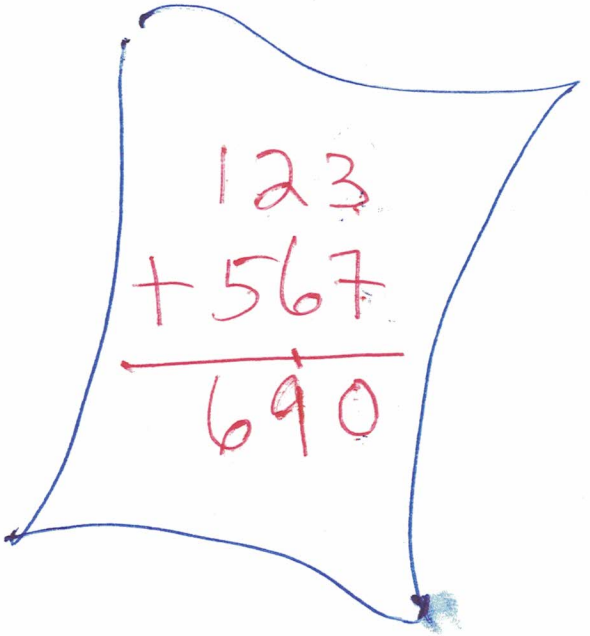
start of C.S.

Alan Turing (1936) at 23.

Father of CS.

Father of AI. (Turing Test)

Turing machine Intuition:

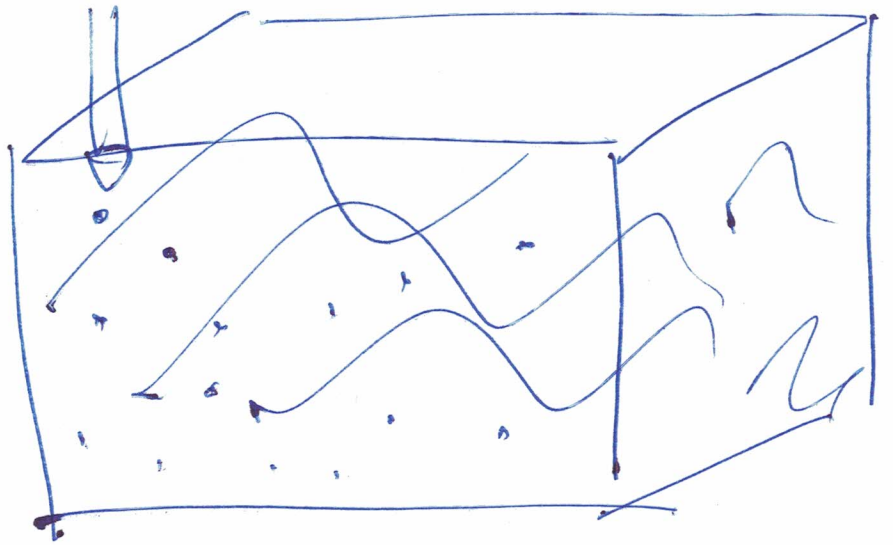


memory { ①. a piece of paper (size = infinite)
②. (local) R/W on the paper.

program { ③. finitely many rules on R/W, move on the paper.

3	1
7	5
0	6

"global" read/write : using chemistry



Biology :

DNA Computing .